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5 July 1973
SUPERSEDING
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28 April 1967

MILITARY SPECIFICATION
COMPONENTS FOR PROPELLANT ACTUATED DEVICES
GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers components of propellant actuated devices used in aircrew emergency escape systems, which are not otherwise covered by a detail specification.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS

Munitions Command

Applicable drawings as listed in the invitation for bids or request for proposal.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1377

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2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute, (ANSI), 1430 Broadway, New York, N.Y. 10018).

ASTM E-18 - Method of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

3. REQUIREMENTS

3.1 General.-Materials, parts, and sub-assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Material.-The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.2.1 Physical or mechanical properties.-Physical or mechanical properties required by the drawing shall be uniform throughout each part within specified tolerances. Mechanical properties for all alloy steel parts shall be achieved by quenching and tempering.

3.3 Threads.-Threads shall be full and undamaged for the entire minimum length as required by the applicable drawings and shall be in accordance with provisions of MIL-A-2550.

3.4 Protective finish.-Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration.

3.5 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specification and the following.

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3.5.1 Metal parts.-The part shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.5.2 Burr.-The part shall not have a burr which might interfere with assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.-The part shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified in the drawings and specifications, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the drawings and specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.2 First article sample.

4.2.1 Initial production sample.-When specified in the contract, an initial production sample shall be submitted at the beginning of regular production in accordance with contract requirements. The samples shall be subjected to examination and tests specified in the applicable drawings, specifications, Supplementary Quality Asurance Provisions (SQAP) and tests in 4.5.4 (when applicable) using the procedures in accordance with 4.6. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility as specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

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4.3 Categories of inspection.-Inspection of items covered by this specification shall be in accordance with one of the following categories of inspection:

Category A - Inspection system
Category B - Statement of findings
Category C - Certification

Unless otherwise specified in the contract, Category A shall apply.

4.3.1 Category A - Inspection system.-The contractor shall provide and maintain an inspection system in accordance with the contract or purchase order.

4.3.1.1 Examination.-Examinations included in the contractor's inspection plan shall include, as a minimum, the requirements of 4.4.

4.3.2 Category B - Statement of findings.-When Category B is authorized in the contract the contractor shall perform such examinations and tests as are required by this specification and other documentation referenced in the contract, and shall submit a statement of findings recording satisfactory compliance with all requirements. A separate statement of findings shall be submitted for each lot of components to be presented for acceptance. If a sub-assembly is being supplied, a separate statement of findings shall be submitted for each lot of such items, plus a statement of findings covering inspection of each lot of the components comprising the sub-assembly.

4.3.2.1 Contents of statement.-Each statement shall include the following data as a minimum:

a. Purchase order or contract number.
b. Item nomenclature and part number.
c. Applicable drawings and specifications, and revision level of each, including supplementary quality assurance provisions, inspection equipment date and packaging data, as applicable.
d. Sampling plan.
1. Lot number and lot size.
2. Sample size, and acceptance criteria (acceptance number and rejection number).
e. Description of each characteristic for which inspection is required by this specification, applicable drawings and other documents referenced in the contract.
f. Method of inspections of each characteristic.
g. Authorized deviations, if any.
h. Statement that inspection has been conducted in accordance with above criteria, and the item has been found to conform with all requirements. Results of inspection shall be shown for each characteristic, including number of rejects, and reason for rejection.
i. Signature of a responsible officer of the contractor.

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4.3.2.2 Government inspection and acceptance.-Items being procured will be accepted upon proper identification with an acceptable Statement of Findings. The Government reserves the right, however, to independently inspect for conformance with all contractual requirements prior to acceptance. The Government further reserves the right to conduct 100 percent inspection and to reject any individual non-conforming items.

4.3.2.3 Supply and maintenance of test equipment.-Unless otherwise specified in the contract, the contractor shall supply all test equipment necessary for performance of inspection requirements of the contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The Government reserves the right to certify inspection equipment, and to use it for Government verification or acceptance inspection, to the extent that such usage by the Government will not unduly affect contractor's production schedules.

4.3.2.4 Examination.-Examination shall be performed in accordance with 4.4.

4.3.3 Category C - Certification.-Where Category C Certification is authorized in the contract, the contractor shall certify that the product being supplied has been determined to be in full compliance with all requirements of the drawings and specifications cited in the contract and that workmanship and materials conform with recognized standards of commercial quality. The certification shall reference the contract or purchase order number, part number, and applicable drawings and specifications together with their revision or amendment numbers and shall be signed by a responisible officer of the contractor.

4.4 Examination under Categroy A and Category B inspection.

4.4.1 Components and assemblies covered by SQAP.-Where SQAP are specified in the contract, the examination and inspection requirements of the SQAP shall represent the contractor's minimum inspection requirements for those components and assemblies to which the SQAP pertain, and shall be the basis of acceptance or rejection by lot. Where a SQAP is specified for an assembly of components, that SQAP, in the absence of any SQAP covering components comprising the assembly, will be construed to cover the minimum inspection requirements for the assembly and its components. In such cases, documented evidence of component inspection beyond that required by the SQAP will not constitute a condition of acceptance by the Government.

4.4.2 Components and assemblies not covered by SQAP.-For components and assemblies not covered by SQAP, all characteristics delineated on drawings shall be subject to examination by the contractor. The minimum amount of inspection shall be determined by the classification of defects in accordance with criteria specified below.

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4.4.2.1 Critical defects.-For purposes of inspection, any defect that judgement and experience indicate is likely to prevent performance of the tactical or life saving function of the end item such as:

a. Missing parts.
b. Improper assembly.
c. X-Ray criteria.

4.4.2.2 Major defects.-For purposes of inspection, any defect in the following types of specified characteristics will be classified as a Major Defect:

a. Linear or diametrical dimensions having tolerances of 0.005 inch or less.
b. Angular and radial dimensions.
c. Concentricity or true position requirements.
d. Screw threads.
e. Mandatory load requirements for springs.
f. Parallelism and perpendicularity requirements.
g. Lubrication requirements.
h. Protective finish.
i. Foreign matter.
j. Burrs.
k. Metal defects.
l. Surface finishes.

4.4.2.3 Minor defects.-Defects in characteristics not listed under 4.4.2.2 shall be classified as Minor defects such as:

a. Location of labels.
b. Scratches.
c. Marking missing incorrect or illegible.
d. Chamfers missing or incorrect.

TABLE I

Lot Size	Sample Size	Accept Numbers Major Accept	Major Reject	Minor Accept	Minor Reject
1 - 300	100% Examination				
301 - 1,300	110	1	2	2	3
1,301 - 3,200	150	2	3	3	4
3,201 - 8,000	225	3	4	4	5
8,000 -22,000	300	4	5	5	6

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TABLE II

Sample Size for Major Defects	Total Allowable Defects Per Sample Size			
	110	150	225	300
No. of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over - 43	6	8	11	14

Sample Size for Minor Defects	Total Allowable Defects Per Sample Size			
	110	150	225	300
No. of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over - 48	8	11	17	20

4.5 Inspection provisions.

4.5.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.5.2 Examination.-Examination shall be performed in accordance with the applicable drawings, specifications and SQAP's and as specified herein. One-hundred percent examination shall be performed for all critical defects, except when tests are destructive. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained in the applicable SQAP, using the sampling plan and associated criteria in Tables I and II. For components and assemblies not covered by SQAP, all characteristics delineated on drawings shall be subject to examination by the contractor. The minimum amount of inspection shall be determined by the classification of defects in accordance with criteria specified in 4.4.2. For an individual defect (single characteristic) the criteria in Table I shall be used. For total defects within a major or minor group, the criteria in Table II shall be used. All non-conforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The roughness comparison specimens prescribed in ANSI B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burrs and foreign matter.

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4.5.3 Packing and marking.-Inspection for packing and marking shall be in accordance with instructions from the procuring activity.

4.5.4 Testing.-The following tests shall be performed in accordance with the provisions of 4.6.

4.5.4.1 Hardness.

4.5.4.1.1 Non-destructive.-When the test is non-destructive, fifty units from each heat treated lot of components shall be tested for compliance with the requirements specified on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components. Lots of fifty components or less shall be tested 100%. Failure of any unit to comply with the drawing requirements shall be cause for rejection of that unit.

4.5.4.1.2 Destructive.-When the test is destructive, the sample size and acceptance criteria shall be as specified in the applicable specification, SQAP or contract.

4.5.4.2 Hydrostatic test.-Each unit shall be subjected to this test. Failure of any unit to comply with the requirements of the applicable drawing, specification or contract shall be cause for rejection of the unit.

4.5.4.3 Shear pin test.-Completely finished shear pins shall be subjected to this test. Fifty (50) shear pins shall be selected at random from each lot. The shear pins shall be subjected to a destructive test by a test laboratory certified by the government to determine compliance with the shear value specified in the applicable drawing. Failure of any pin to meet the requirements of the applicable drawing shall be cause for rejection of the lot.

4.5.4.4 Protective finish.-The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable Military Standard listed on the drawing.

4.6 Test methods and procedures.

4.6.1 Hardness test.-Prior to application of protective finish, a hardness test shall be performed. The tests shall be performed using the equipment and procedures for performing this test as prescribed by ASTM E-18.

4.6.2 Hydrostatic test.-The unit shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. Each unit shall be individually mounted in a test device using the fixture for sealing all openings at the prescribed seal surfaces without applying an axial load other

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4.6.2 (Cont'd)

than that applied by the liquid pressure. In performing the test, the pressure shall be applied gradually until the required minimum pressure is reached, after which the pressure shall be held for a minimum of fifteen (15) seconds or until the minimum time specified on the drawing or the specification. No extraneous forces by the holder, adapter or equipment on the unit tested is allowable. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over two (2) hours of continuous operation. It shall also be calibrated prior to restarting the operation after a layoff of one (1) hour or more.

4.6.3 Shear pin test.-The shear pins shall be tested for shear value by using test equipment and fixtures capable of performing the test accurately and consistently. All data must be recorded and the results forwarded with the lot.

4.6.4 Protective finish.-Test procedures for protective finish shall be in accordance with the applicable Military Standard listed on the drawing.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-Unless otherwise directed by the procuring activity, packaging, packing and marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 The contractor shall be responsible for design, fabrication, calibration and maintenance of all inspection gages, test equipment and standards necessary to assure the government a quality item.

6.2 Definitions.

6.2.1 SQAP - (Supplementary Quality Assurance Provisions.- A SQAP is an engineering document which prescribes minimum contractor inspection requirements and assembly is identified by the drawing number of the item prefixed by "SQAP".

6.2.2 Abbreviations.-SME and STE appearing on SQAP forming a part of this specification are defined as follows:

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6.2.2.1 Standard Measuring Equipment(SME)-Standard Measuring Equipment (SME) is defined as the common hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.2.2.2 Special Test Equipment(STE).-Special Testing Equipment is defined as special inspection fixtures, instruments, and modified standard measuring equipment. Appearance of "STE" on a SQAP does not imply that the equipment is available, or that a suitable design for such equipment is currently in existance.

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Preparing activity:

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